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Academic year 2024-2025 (Even Sem) CIE - 1 - 2025

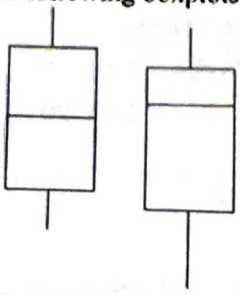
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**01/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM241AT**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I**	
STATISTICS FOR DATA ANALYTICS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	40th percentile in a given set of 140 data values occurs at position _____	(1)	L2	CO1
2	<table><tr><td>Class</td><td>21"-30"</td><td>31"-40"</td><td>41"-50"</td><td>51"-60"</td><td>61"-70"</td></tr><tr><td>No of occurrences</td><td>12</td><td>16</td><td>22</td><td>25</td><td>18</td></tr></table> The median in the above data lies in the class _____	(1)	L2	CO1
3	According to Bayes' rule, P(A|B) = ___, for two events in a sample space S.	(1)	L1	CO1
4	X = the no. coming up when a 6-faced die is thrown. Y = no. of heads when a coin is tossed. The possible values of X - Y are _____.	(1)	L3	CO2
5	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A′∩ B) = ___.	(1)	L3	CO2
6	P(X <= 3| Y=0) is ___, for X and Y as in Question no. 1.4	(1)	L3	CO2
7	A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective? _____	(2)	L3	CO1
8	Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: <table><tr><th rowspan="2">Type of unsaturation</th><th colspan="2">Type of oil</th></tr><tr><th>Canola</th><th>Corn</th></tr><tr><td>Mono</td><td>7</td><td>13</td></tr><tr><td>Poly</td><td>93</td><td>77</td></tr></table> If a bottle of oil is selected at random, what is the probability that it belongs to the polyunsaturated category?	(2)	L3	CO1
	Part –B			
1	Fourteen measurements of furnace temperatures (in ^{0}F) are recorded on successive batches in a semiconductor manufacturing process as follows: 953, 950, 948, 955, 951, 949, 957, 954, 955, 949, 952, 931, 953, 955. a) Calculate sample mean, sample variance, and standard deviation. b) Find the mode and median. How much could the largest temperature measurement be increased without changing the median value? c) Construct a box plot of the data, showing clearly the outlier bounds. d) Comment on the symmetry of the data.	(10)	L3	CO1
2 a	Describe any similarities or differences in the two distributions represented by	(05)	L4	CO1

Academic year 2024-2025 (Even Sem)

	the following boxplots. Assume the two boxplots have the same scale. 			
b	In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1,2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins what is the probability that it is defective and find the conditional probability that it was produced by machine 3	(05)	L4	CO1
3	Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products.			
	a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review?	(04)	L3	CO2
	b) If a product attains a good review, what is the probability that it will be a highly successful product?	(03)	L3	CO2
	c) If a product does not attain a good review, what is the probability that it will be a highly successful product?	(03)	L3	CO2
4	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G)	(10)	L3	CO2
5	Suppose the excess length in a manufactured part in mm is recorded as below.	(10)	L2	CO1

19	22	6	29	57	14	12	24	6	37
19	19	18	12	11	17	15	17	13	56

i. Compute the mean, median and mode for the data set.
ii. Draw a histogram with 4 classes and a box plot, and interpret.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	7	3	-	-	1	2	7	-	-	-
	Test		30	20	-	-	-	10	20	30	-	-

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DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

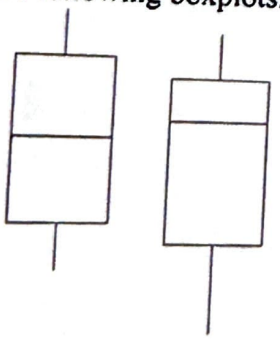
Date	**01/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM241AT**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I**	
STATISTICS FOR DATA ANALYTICS (scheme and solution)			

Note:

1. **Answer all the Questions.**

Sl. No.	Questions	M	BT	CO
	Part – A			
1	40th percentile in a given set of 140 data values occurs at position ___56th_____ 40th percentile in 140 data values • Position = (40/100) × 140 = 56th • Answer: 56th position	(1)	L2	CO1
2	<table><tr><td>Class</td><td>21"-30"</td><td>31"-40"</td><td>41"-50"</td><td>51"-60"</td><td>61"-70"</td></tr><tr><td>No of occurrences</td><td>12</td><td>16</td><td>22</td><td>25</td><td>18</td></tr></table>The median in the above data lies in the class __41 -50'' class_____ Cumulative frequencies: 12, 28, 50, 75, 93 → Median is at 47th position Answer: 41"-50"	(1)	L2	CO1
3	According to Bayes' rule, P(A|B) = ___, for two events in a sample space S. P(A|B)=P(B)*P(B|A) / P(A)	(1)	L1	CO1
4	X = the no. coming up when a 6-faced die is thrown. Y = no. of heads when a coin is tossed. The possible values of X - Y are _0 to 6_____.	(1)	L3	CO2
5	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A′∩ B) =__ S=A∪B∪C∪D={1,2,3,4,5,6,7,8,9}, total 9 A′∩B={6,8}A' ∩ B = \{6,8\}A′∩B={6,8} → P = 2/9.	(1)	L3	CO2
6	P(X <= 3| Y= 0) is ___ , for X and Y as in Question no. 1.4 When Y=0 → Coin = T, X=1 to 6 X ≤ 3: 1,2,3 → 3 favourable out of 6 **Ans: 36=0.5\frac{3}{6} = 0.563=0.5**	(1)	L3	CO2
7	A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective? _____ Without replacement: (25/100)×(24/99)=(600/9900) ≈0.0606	(2)	L3	CO1
8	Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: <table><tr><td rowspan="2">Type of unsaturation</td><td colspan="2">Type of oil</td></tr><tr><td>Canola</td><td>Corn</td></tr><tr><td>Mono</td><td>7</td><td>13</td></tr><tr><td>Poly</td><td>93</td><td>77</td></tr></table>If a bottle of oil is selected at random, what is the probability that it belongs to	(2)	L3	CO1

Academic year 2024-2025 (Even Sem)

	the polyunsaturated category? Total =190 Prob =170/190 =17/19 = 0.8947			
1	Part –B Fourteen measurements of furnace temperatures (in ^{0}F) are recorded on successive batches in a semiconductor manufacturing process as follows: 953, 950, 948, 955, 951, 949, 957, 954, 955, 949, 952, 931, 953, 955. a) Calculate sample mean, sample variance, and standard deviation. b) Find the mode and median. How much could the largest temperature measurement be increased without changing the median value? c) Construct a box plot of the data, showing clearly the outlier bounds. d) Comment on the symmetry of the data. Mean = (sum of 14 values)/14 = 13342 / 14 = **952.57°F** Median = 952 Mode = 955 Sample variance s2s^2s2 = approx. **43.45** Standard deviation ≈ **6.59** Largest temperature that doesn't change median = 955 (since 7th and 8th values are 953, 955) Boxplot: • Outliers = values outside Q1–1.5IQR or Q3+1.5IQR • 931 is an outlier Symmetry: Slight left skew due to low outlier (931)	(10)	L3	CO1
2 a	Describe any similarities or differences in the two distributions represented by the following boxplots. Assume the two boxplots have the same scale.  Look at median (central tendency), IQR (spread), symmetry (skewness) Eg: Boxplot A has longer right whisker → positive skew	(05)	L4	CO1
b	In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1,2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins what is the probability that it is defective and find the conditional probability that it was produced by machine 3 P(D)=0.35×0.02+0.25×0.01+0.40×0.03=0.007+0.0025+0.012=0.0215 P(M3|D) =0.40×0.030.0215≈0.558P	(05)	L4	CO1

3	Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products. a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review? b) If a product attains a good review, what is the probability that it will be a highly successful product? c) If a product does not attain a good review, what is the probability that it will be a highly successful product? P(Good Review) = 0.4×0.95 + 0.35×0.6 + 0.25×0.1 = 0.38 + 0.21 + 0.025 = **0.615** P(Highly Successful | Good) = 0.38 / 0.615 ≈ **0.617** P(Highly Successful | Not Good) = 0.02 / 0.385 ≈ **0.052**	(04) (03) (03)	L3 L3 L3	CO2 CO2 CO2
4	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G) Total = 10 P(D|C) = 3/4 = 0.75 P(D|G) = 2/6 = 1/3 P(G|D) = 2/5 = 0.4 P(C|S) = 1/5 = 0.2 P(G) = 6/10 = 0.6	(10)	L3	CO2
5	Suppose the excess length in a manufactured part in mm is recorded as below. 19 22 6 29 57 14 12 24 6 37 19 19 18 12 11 17 15 17 13 56 i. Compute the mean, median and mode for the data set. ii. Draw a histogram with 4 classes and a box plot, and interpret. • Mean ≈ 22.15 mm • Median = 17 • Mode = 19 • Histogram: 4 classes → 0–15, 16–30, 31–45, 46–60 • Box plot: Outliers likely at 56, 57 • Interpretation: Positive skew due to high values	(10)	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	7	3	-	-	1	2	7	-	-	-
	Test		30	20	-	-	-	10	20	30	-	-

DEPARTMENT OF MECHANICAL ENGINEERING			
CONTINUOUS INTERNAL EVALUATION (CIE) – Test 1			
Date	**02/04/2025**	**Maximum Marks**	**10+50**
Course Code	**ME242TB**	**Duration**	**120 Min**
Sem	**IV Sem**	**Academic Year**	**2024-25 (Even Sem)**
Course – MATERIAL SCIENCE FOR ENGINEERS			

#		PART A	Marks	BT	CO
1		According to which principle the electrons fill the lowest energy orbital first before moving to higher energy ones.	1	1	1
2		In _____ material the conduction and valence bands overlap so electrons can easily pass into the conduction band.	1	1	1
3		Define unit cell.	1	1	1
4		Write the classification of composites based on matrix.	2	1	1
5		The Burgers Vector is parallel to the dislocation line, the type of line defect is _____.	1	1	1
6		If an atom is missing from its normal site in the matrix, the defect is called _____ defect.	1	1	2
7		Define thermal conductivity with equation.	2	2	2
8		In dielectric materials the movement of electrons in response to an external electric field is _____type of polarization mechanism.	1	1	2
#		PART B	Marks	BT	CO
1	a	With the help of neat sketches explain the Primary Interatomic bonds with examples.	10	2	1
2	a	Classify solid materials explain them briefly with examples	10	2	1
3	a	Calculate Atomic packing factor for FCC unit cell	05	3	2
	b	Classify and explain different crystal imperfections	05	2	2
4	a	Compare the properties and applications of metals and polymers	10	3	1
5	a	Illustrate the following thermoelectric effects i) Seebeck effect ii) Peltier effect	05	3	2
	b	Discuss different types of Insulating materials highlighting their applications	05	2	2

Course Outcomes:	
CO 1	Understand the classification of materials, their atomic structure, and properties.
CO 2	Investigate the properties and applications of different materials.
CO 3	Analyze the effect of different heat treatment processes.
CO 4	Recognize different types of nanomaterials, synthesis methods and characterization techniques.

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2024-2025 (Even Sem)

Date	2nd April 2025	Maximum Marks	60
Course Code	BT242TC	Duration	120 min
Sem	IV Semester	Closed Book Offline Quiz & Test -1	

Biosafety Standards and Ethics (Basket Course)

Sl. No	Questions (Quiz)	M	BT-L	CO
1.	What is the role of the Institutional Biosafety Committee (IBC) in India?	1	2	2
2.	The Recombinant DNA Advisory Committee (RCGM) is a part of which Department of Govt of India?	1	1	2
3.	The committee of an Institutional Biosafety Committee (IBC) comprise of _____ and _____	1	1	2
4.	Biosafety levels indicate _____	1	1	1
5.	_____ is the primary mode of transmission of many viral biohazards.	1	1	1
6.	Which PPE is essential for BSL 4 labs?	1	2	1
7.	In class II biosafety cabinet, which direction does the air flow	1	2	2
8.	The Cartagena Protocol was adopted in which year?	1	1	2
9.	Mention the 3 types of mechanism which takes place in HEPA filters	2	3	2

Sl. No	Questions (Test)	M	BT-L	CO
1.	Comment on various types of Biohazards, and their impact on human health.	10	2	1
2.	Explain Biosafety level II class B2 with a neat diagram	10	2	1
3.	Discuss the importance of biosafety guidelines in India and how they contribute to public health and safety.	10	3	2
4.	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	10	3	2
5.	How do the Biosafety guidelines of the Government of India align with international biosafety standards? Discuss any significant differences or similarities.	10	2	2

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	24	36	---	---	5	33	22	---	---	---

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Department of Civil Engineering

Date	02/04/2025	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min +20 Min
Sem	III	CIE – I (Quiz + Test)	
Environment and Sustainability (Basket Course)			

Part A

Sl. No.	Questions	M	BT	CO
1.	List the components of environment.	2	1	1
2.	Define biodiversity.	2	2	1
3.	Differentiate between primary and secondary pollutants.	2	3	2
4.	The permissible day and night noise levels for a silent zone in India are dB and dB respectively.	2	2	2
5.	List any four sources of solid waste.	2	2	1

Part B

Sl. No.	Questions	M	BT	CO
1.	Classify different types of eco system and explain any two briefly.	10	2	1
2.	Define solid waste management and list the objectives of solid waste management.	10	2	2
3.	Briefly explain the following safety terminologies. i) Accident ii) Ergonomics iii) First aid iv) Hazard v) Safety inspection	10	2	2
4.	Define noise pollution. List the effects of noise pollution and write the control measures.	10	2	2
5.	With a neat sketch explain carbon cycle.	10	2	1

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	26	34	-	-	2	46	2	-	-	-

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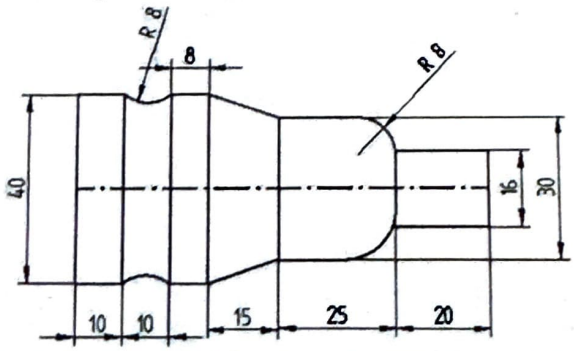
Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	1st April 2025	Maximum Marks	10 + 50
Course Code	IM343AI	Duration	120 Min
Sem	IV Semester	CIE –I	
CAD/CAM & ROBOTICS			

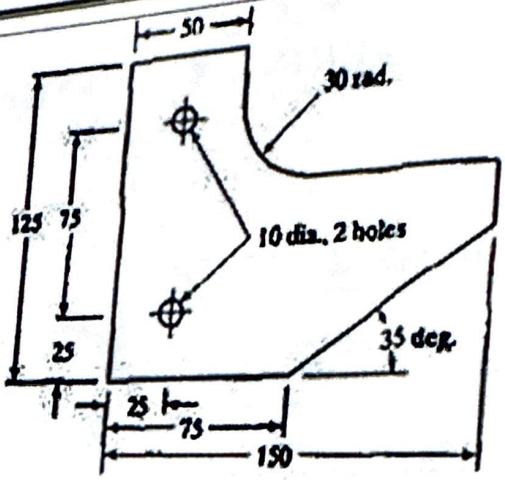
Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1.1	List any two advantage of using CAD over manual drafting?	02	L2	CO1
1.2	How does DNC differ from CNC? State any two.	02	L1	CO2
1.3	List any two reasons for implementing CAD in engineering and product design?	02	L2	CO2
1.4	What are the key components of a DNC system?	02	L2	CO1
1.5	What are two essential elements of an NC system?	02	L1	CO2
	Part –B			
1	Discuss the various applications of computers in the design process with its various stages and their importance in engineering design.	10	L1	CO2
2	According to the type of machine, NC machine are classified into three types explain the classification with the help of Case examples.	10	L2	CO1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes 	10	L4	CO2
4	Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool-Slot drill Cutter of ϕ10mm, Cutting Speed-1500, Feed- 60, Depth of cut=6mm	20	L4	CO1

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Academic year 2024-2025 (EVEN Sem)



BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Ma x Marks	04	06	-	-	04	06	-	-	-	--
	Test		20	30	--	-	10	10	-	30	-	--

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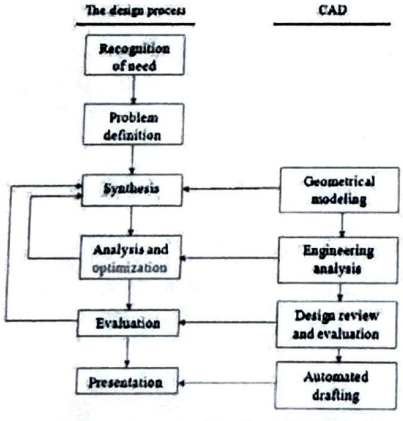
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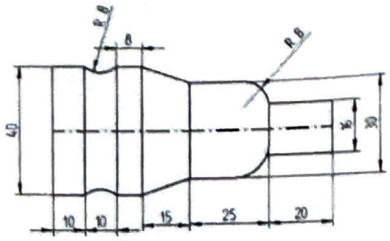
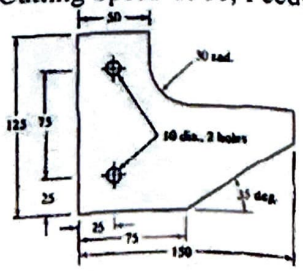
Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	1st April 2025	Maximum Marks	10 + 50
Course Code	IM343AI	Duration	120 Min
Sem	IV Semester	CIE –I	
CAD/CAM & ROBOTICS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M
	Part – A	
1.1	• CAD allows precise measurements and error-free drawings. • Designs can be edited or updated quickly without redrawing the entire layout.	02
1.2	• CNC (Computer Numerical Control) controls a single machine, while DNC (Direct Numerical Control) can control multiple machines from a central computer. • In CNC, programs are loaded directly onto the machine; in DNC, the central computer sends instructions directly to machines in real-time.	02
1.3	• Speeds up the design process and reduces development time. • Offers 2D and 3D views, helping in better understanding and communication of designs.	02
1.4	• Central Computer: Manages and distributes part programs to multiple machines. • Communication Lines/Networks: Connect the central computer to CNC machines for data transmission.	02
1.5	• Program of Instructions: The coded data that defines operations (often in G-code). • Machine Control Unit (MCU): Interprets the instructions and controls the machine's movements.	02
	Part –B	
1	Discuss the various applications of computers in the design process with its various stages and their importance in engineering design.  There are four steps in the CAD design process: • Geometric modeling (in place of synthetic step): This is a mathematical technique that is done by using simulation software. It shows the object from the math formula through computer graphics. • Engineering analysis (in place of analysis and optimization): This is the process of checking the strength of moving objects to determine how they will work in the real world.	10

	• Design review and evaluation (replaces evaluation): Design is reviewed and evaluated with the help of CAD software. Also, the feasibility of the design is checked. • Automatic drafting (replaces presentation): The design is checked to see if it meets product specifications. The design is then saved automatically.	10
2	The classification of NC machine tool systems can be done in three ways: 1. According to the type of machine: Point-to-point, Straight cut and contouring (continuous path) 2. According to the programming method: incremental versus absolute 3. According to the type of control loops: Open-loop versus closed-loop	10
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes  G21 G98 G28 U0 W0 M06 T01 M03 S1000 G00 X30 Z05 G71 U1 R0.5 G71 P10 Q20 U0 W0 F80 N10 G01 X16 F20 G01 Z-20 G03 X30 Z-28 R8 G01 X30 Z-45 G01 X40 Z-60 G01 X40 Z-68 N20 G03 X40 Z-78 R8 G01 X42 G28 U0 W0 M05 M30	10
4	Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool-Slot drill Cutter of ϕ10mm, Cutting Speed-1500, Feed- 60, Depth of cut=6mm  Post processor-MACHIN/MILL,1	20

Spindle speed – 1500rpm
Feed rate – 60mm/min
Tolerance – 0.01mm/min
Thickness- 15mm
Cutter dia – 20mm
Cutter location - -50.0, -50.0, 50.0

PART NO/1	SPINDL/1500
MACHIN/MILL,01	FEDRAT/60
INTOL/0.01	COOLANT/ON
OUTOL/0.01	FROM/P0
CUTTER/20.0	GO/TO, L1 TO PL1, TO P5
SPINDL/1000	GOLFT/L1,PAST, P2
FEDRAT/75	GORGT/P2, PAST, L2
COOLNT/ON	GOFWD/L2, PAST, P3
STRT=POINT/-50.0, -50.0, 50.0	GODWN/L3, PAST, L4
P0=POINT/0.0,0.0, 0.0	GOFWD/C1, PAST, P5
P1=POINT/0,0, -6	GORGT/L4, PAST, P6
P2=POINT/0,125, -6	GORGT/L5,PAST, P7
P3=POINT/50,125, -6	GORGT/L6, PAST, P8
P4=POINT/50,100, -6	GORGT/L7, PAST, P1
P5=POINT/75,75, -6	GOTO/P0
P6=POINT/150,75, -6	SPINDL/OFF
P7=POINT/150,52.5, -6	COOLNT/OFF
P8= POINT/75,0, -6	FINI
P9=POINT/80,100, -6	
L1=LINE/P1, P2	
L2=LINE/P2,P3	
L3=LINE/P3, P4	
L4=LINE/P5, P6	
L5=LINE/P6, P7	
L6=LINE/P7, P8	
L5=LINE/P8, P1	
C1=CIRCLE/CENTER, P9, RADIUS, 30	
PL1=PLANE/P1, P2, P3	

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Academic year 2024-2025 (EVEN Sem)

EPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	02nd April 2025	Maximum Marks	10 + 50
Course Code	IM244AI	Duration	30 + 90 Min
Sem	IV	CIE – I	

OPERATIONS RESEARCH

NOTE:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	The variables that represent decisions to be made in an optimization model are called _____.	1	1	1
2	In an optimization model, the function that needs to be maximized or minimized is called the _____.	1	1	1
3	Linear Programming is a mathematical technique used for _____ decision-making.	1	2	2
4	One of the key limitations of Operations Research is that it requires _____ data for accuracy.	1	2	2
5	The feasible region is the set of all possible solutions that satisfy all _____.	1	1	1
6	The graphical solution method of LPP is applicable only when there are _____ decision variables.	1	2	2
7	The additivity assumption ensures that the total effect of different decision variables is the _____ of their individual effects.	1	2	1
8	When the total supply does not equal total demand, the transportation problem is called a(n) _____ transportation problem.	1	2	2
9	A prohibited route transportation problem occurs when certain routes have _____ restrictions and cannot be used for transportation.	1	2	3
10	The General Mathematical Formulation of a transportation problem consists of an objective function, supply constraints, and _____ constraints.	1	2	2
	Part –B			
1	Describe the Seven-Step Model-Building Process in Operations Research with an example.	10	2	1
2	Explain the key features of Operations Research (OR) models and highlight its limitations.	10	2	1
3	Consider the following problem faced by a production planner of a soft drink plant. He has two bottling machines A and B. A is designed for 8-ounce bottles and B for 16-ounce bottles. However, each can also be used for both types of bottles with some loss of efficiency. The manufacturing data is as follows: **Machine** | **8-ounce Bottles** | **16-ounce Bottles** A | 100/minute | 40/minute B | 60/minute | 75/minute The machines can be run for 8 hours per day, 5 days per week. The profit on an	10	3	2

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Academic year 2024-2025 (EVEN Sem)

	8-ounce bottle is Rs 1.5 and on a 16-ounce bottle is Rs 2.5. Weekly production of the drink cannot exceed 3,00,000 bottles and the market can absorb 25,000, 8-ounce bottles and 7,000, 16-ounce bottles per week. The planner wishes to maximize his profit, subject of course, to all the production and marketing restrictions. Formulate this problem as an LP model to maximize total profit.			
4	A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals and 1,400 calories. Two foods A and B are available at a cost of Rs. 4 and Rs. 3 per unit, respectively. If one of A contains 200 units of vitamins, 1 unit of mineral and 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 calories. Formulate this problem as an LP model and solve it by graphical method to find combination of foods to be used to have least cost.	10	3	2
5	Determine an initial basic feasible solution to the following transportation problem by using (a) NWCR, (b) LCM and (c) VAM.	10	3	3

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	5	1	-	3	7	-	-	-	-
	Test		20	20	10	-	-	20	30	-	-	-

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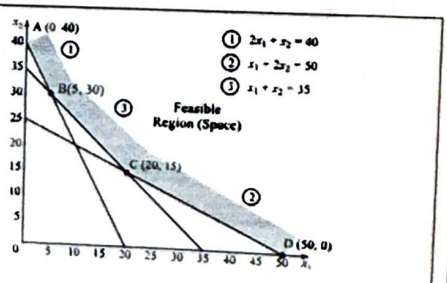
Academic year 2024-2025 (ODD Sem)

EPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**02/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM244AI**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I-Scheme**	
Operations Research			

Note:
1. Answer all the Questions.

Sl. No.	**Questions**	**M**
	Part – A	
1	decision variables	1
2	objective function	1
3	optimization	1
4	accurate	1
5	constraints	1
6	two	1
7	sum	1
8	unbalanced	1
9	zero/infinite cost	1
10	demand	1
	Part –B	
1	The Seven-Step Model-Building Process is used to structure and solve decision-making problems systematically. 1. Problem Definition 2. Data Collection 3. Model Formulation 4. Model Solution 5. Model Validation 6. Implementation 7. Monitoring & Updating Listing-3M,Explanation-7M	10
2	**Features of OR Models:** Scientific Approach, Optimization Focus ,Interdisciplinary Nature ,Quantitative Decision-Making, Systems Approach, Simulation & Prediction , Iterative Process ---5M **Limitations of OR**: Complexity, Data Dependency ,High Computational Requirements, Time-Consuming ,Assumptions vs. Reality, Implementation Challenges, Not Always Practical--------5M	10

3	**LP model formulation** The data of the problem is summarized as follows: | Constraints	Production		Availability | |---|---|---|---| |	8-ounce Bottles	16-ounce Bottles	| | Machine A time	100/minute	40/minute	$8 \times 5 \times 60 = 2{,}400$ minutes | | Machine B time	60/minute	75/minute	$8 \times 5 \times 60 = 2{,}400$ minutes | | Production	1	1	3,00,000 units/week | | Marketing	1	–	25,000 units/week | |	–	1	7,000 units/week | | Profit/unit (Rs)	1.5	2.5	| ***Decision variables*** Let x_1 and x_2 = units of 8-ounce and 16-ounce bottles to be produced weekly, respectively ***The LP model*** Maximize (total profit) $Z = 1.5x_1 + 2.5x_2$ **subject to the constraints** (i) Machine time : (a) $\frac{x_1}{100} + \frac{x_2}{40} \le 2{,}400$ and (b) $\frac{x_1}{60} + \frac{x_2}{75} \le 2{,}400$ (ii) Production : $x_1 + x_2 \le 3{,}00{,}000$ (iii) Marketing : (a) $x_1 \le 25{,}000$, (b) $x_2 \le 7{,}000$ **and** $x_1, x_2 \ge 0$. Table(4M)+decision variables and objective(2M)+constraints(4M)	10
4	**Solution** The data of the given problem can be summarized as shown below. | Food | Units Content of | | | Cost per Unit (Rs.) | |---|---|---|---|---| | | Vitamins | Mineral | Calories | | | A | 200 | 1 | 40 | 4 | | B | 100 | 2 | 40 | 3 | | Minimum requirement | 4,000 | 50 | 1,400 | | Let us define the following decision variables: x_1 and x_2 = number of units of food A and B to be used, respectively. Then LP model of the given problem is: Minimize (total cost) $Z = 4x_1 + 3x_2$ subject to the constraints (i) Vitamins : $200x_1 + 100x_2 \ge 4{,}000$, (ii) Mineral : $x_1 + 2x_2 \ge 50$, (iii) Calories : $40x_1 + 40x_2 \ge 1{,}400$ and $x_1, x_2 \ge 0$ Formulation-4M  Graph-3M,Answer-3M	10
5	NWC:Total cost = $5 \times 19 + 2 \times 30 + 6 \times 30 + 3 \times 40 + 4 \times 70 + 14 \times 20$ = Rs 1,015------3M LCM:Total cost = $7 \times 10 + 2 \times 70 + 7 \times 40 + 3 \times 40 + 8 \times 8 + 7 \times 20$ = Rs 814-------3M VAM: Total cost = $5 \times 19 + 2 \times 10 + 7 \times 40 + 2 \times 60 + 8 \times 8 + 10 \times 20$ = Rs 779—4M	10

USN

Academic year 2024-2025(Even Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	3rd April 2025	Maximum Marks	10+50
Course Code	IM345AT	Duration	120 Min
Sem	IV	CIE - I	

MARKETING MANAGEMENT

PART – A

Sl. No.	Questions	M	BT	CO
1	Differentiate between marketplace and market space.	02	L2	1
2	What is the mantra of marketing?	02	L2	1
3	Define affiliate marketing and give an example.	02	L1	2
4	Mention any four characteristics of a good content.	02	L1	1
5	Differentiate between push and pull marketing.	02	L2	2

PART – B

Sl. No.	Questions	M	BT	CO
1.	A newly launched e-commerce company wants to build brand awareness through content marketing. Suggest a step-by-step strategy for them, including content types and distribution channels.	10	L2	3
2.	A travel agency specializing in adventure tourism wants to improve its targeted marketing. How should they create a buyer persona, and what factors (e.g., demographics, interests, online behaviour) should they consider?	10	L4	3
3.	A retail company has been using traditional marketing methods such as newspaper ads, billboards, and TV commercials. However, they are now considering a shift to digital marketing. Compare the benefits and limitations of both approaches and suggest a strategy for transitioning to digital marketing	10	L4	1
4.	Identify and explain the various entities that can be marketed, providing relevant examples for each category."	10	L3	2
5.	How have the 4Ps of marketing (Product, Price, Place, and Promotion) evolved in the digital era? Provide examples of companies effectively leveraging digital strategies for each element	10	L3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	--	--	--	04	06	--	--	--	--
		Test	10	10	20	10	--		30	10	10	--	--

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	7 APRIL 2025	Maximum Marks	50
Course Code	IM345AT	Duration	90 Min
Sem	IV Semester	Test-1	

MARKETING MANAGEMENT- Scheme of Valuation

Sl. No.	Questions	M	BT	CO
1	Differentiate between marketplace and market space.	02	L2	1
A	**Marketplace:** A **physical or digital location** where buyers and sellers come together to exchange goods and services. **Market Space:** A **conceptual or virtual environment** where commercial value is created through digital networks and communication technologies.			
2	What is the mantra of marketing?	02	L2	1
A	Create, Communicate, Deliver, Value, Target, Profit			
3	Define affiliate marketing and give an example.	02	L1	2
A	**Affiliate marketing** is a type of **performance-based marketing** where a business rewards **affiliates** (partners or individuals) for **driving traffic, leads, or sales** to the company's products or services through the affiliate's **marketing efforts.**			
4	Mention any four characteristics of a good content.	02	L1	1
A	Credible, shareable, useful, interesting, relevant, different and on brand			
5	Differentiate between push and pull marketing.	02	L2	2
A	**Push Marketing:** • **Definition:** Promotes products by **pushing** them towards customers. **Pull Marketing:** • **Definition:** Encourages customers to **actively seek out** the product.			

Sl. No.	Questions	M	BT	CO
1.	A newly launched e-commerce company wants to build brand awareness through content marketing. Suggest a step-by-step strategy for them, including content types and distribution channels.	10	L2	3
A	Goal setting, audience mapping, content ideation and planning, content creation, content distribution, content amplifier, content marketing evaluation, content marketing improvement			
2.	A travel agency specializing in adventure tourism wants to improve its targeted marketing. How should they create a buyer persona, and what factors (e.g., demographics, interests, online behaviour) should they consider?	10	L4	3
A	**Steps to Create a Buyer Persona:** 1. **Research Your Current Customers:** o Analyse your **best existing clients** — those who book often, leave reviews, and engage with your content. o Conduct **surveys or interviews** (online or post-trip feedback forms). 2. **Segment Your Audience:**			

	○ Divide your audience into categories based on travel habits, destination preferences, or trip budgets. 3. **Gather Data (Use Tools like Google Analytics, Social Media Insights):** ○ Identify patterns in **who interacts** with your website and social pages. 4. **Build the Persona:** ○ Give it a **name**, photo, and backstory to make it relatable (e.g., *"Adventurous Ananya"* or *"Explorer Ethan"*). ○ Include key data points			
3.	A retail company has been using traditional marketing methods such as newspaper ads, billboards, and TV commercials. However, they are now considering a shift to digital marketing. Compare the benefits and limitations of both approaches and suggest a strategy for transitioning to digital marketing	10	L3	1
A	**Traditional vs. Digital Marketing** **Aspect** — **Traditional Marketing** — **Digital Marketing** **Channels** — TV, radio, newspaper, magazines, billboards — Websites, social media, email, SEO, PPC, influencer marketing **Reach** — Local or mass (depending on medium) — Global and highly targeted **Cost** — Often expensive (TV, print ads) — Cost-effective (pay-per-click, organic reach options) **Interactivity** — One-way communication — Two-way, real-time engagement with customers **Measurability** — Hard to track ROI accurately — Easily trackable with analytics and KPIs **Flexibility** — Less flexible once launched — Highly flexible and adjustable in real time **Time to Launch** — Requires more time for planning and production — Quicker to launch and adapt			
4.	Identify and explain the various entities that can be marketed, providing relevant examples for each category."	10	L2	2
A	Goods, Services, Experiences, Events, Persons, Places, Properties, Organizations, Information and Ideas			
5.	How have the 4Ps of marketing (Product, Price, Place, and Promotion) evolved in the digital era? Provide examples of companies effectively leveraging digital strategies for each element	10	L2	4
A	**Product → Digital & Personalized Products** • **Evolution:** Products are now more **customizable, data-driven,** and include **digital formats** (e.g., software, online services, virtual goods). • **Example:** **Spotify** uses user data to create personalized playlists, enhancing the product experience. **Price → Dynamic & Transparent Pricing** • **Evolution:** Real-time pricing, discounts, and **personalized offers** are common. Customers can easily compare prices online. • **Example:** **Amazon** uses AI to dynamically adjust prices based on demand,			

	competition, and user behavior. **Place → Online Channels & Omnichannel Presence** • **Evolution:** Shift from physical stores to **e-commerce, apps,** and **marketplaces**. Customers expect seamless experiences across online and offline platforms. • **Example:** **Nike** offers a smooth shopping experience across its website, mobile app, and physical stores (click-and-collect, store locator, etc.). **Promotion → Digital & Interactive Marketing** • **Evolution:** Promotion has shifted from mass media to **targeted digital campaigns, social media, influencer marketing,** and **content marketing**. • **Example:** **Coca-Cola** runs engaging social media campaigns and interactive content (e.g., "Share a Coke") across digital platforms.			

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18.06.2024**	**Maximum Marks**	**50**
Course Code	**IM241AT**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.

<table>
<tr><th>Sl. No.</th><th>Questions</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>1</td><td>Calculate Mean, Mode and Median for following group data
<table>
<tr><th>Class Interval</th><th>Frequency</th></tr>
<tr><td>1 under 3</td><td>4</td></tr>
<tr><td>3 under 5</td><td>12</td></tr>
<tr><td>5 under 7</td><td>13</td></tr>
<tr><td>7 under 9</td><td>19</td></tr>
<tr><td>9 under 11</td><td>7</td></tr>
<tr><td>11 under 13</td><td>5</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>2</td><td>Construct a box plot for following data:
<table>
<tr><td>87</td><td>85</td><td>84</td><td>84</td><td>82</td><td>82</td><td>82</td><td>81</td><td>81</td><td>81</td></tr>
<tr><td>80</td><td>79</td><td>79</td><td>77</td><td>76</td><td>75</td><td>74</td><td>74</td><td>74</td><td>73</td></tr>
<tr><td>73</td><td>73</td><td>73</td><td>72</td><td>72</td><td>71</td><td>71</td><td>71</td><td>70</td><td>69</td></tr>
<tr><td>69</td><td>68</td><td>68</td><td>65</td><td>65</td><td>64</td><td>64</td><td>63</td><td>62</td><td>62</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>3</td><td>Assume the following are the marks obtained in Statistics for Data Analytics CIE1. Calculate Mean, Median, Mode, Variance and Standard Deviation for the data.
<table>
<tr><td>33</td><td>43</td><td>44</td><td>45</td><td>46</td><td>49</td><td>31</td><td>32</td><td>34</td><td>36</td></tr>
<tr><td>39</td><td>26</td><td>29</td><td>22</td><td>22</td><td>23</td><td>25</td><td>11</td><td>19</td><td>18</td></tr>
<tr><td>19</td><td>15</td><td>23</td><td>45</td><td>44</td><td>29</td><td>39</td><td>38</td><td>28</td><td>27</td></tr>
<tr><td>26</td><td>28</td><td>35</td><td>39</td><td>38</td><td>38</td><td>37</td><td>37</td><td>28</td><td>29</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>4</td><td>Construct a Histogram for the following data.
35, 339, 650, 864, 1025, 1261, 38, 340, 655, 883, 1028, 1280, 48, 395, 669, 883 1036, 1290, 53, 457, 703, 890, 1044, 1312, 70, 478, 730, 934, 1087, 1341, 99 501, 763, 951, 1091, 1355, 138, 521, 788, 969, 1126, 1357, 164, 556, 789, 985 1176, 1360, 220, 583, 789, 993, 1199, 1414, 265, 595, 802, 997, 1199, 1436, 272, 596, 822, 999, 1237, 1479, 312, 604, 851, 1018, 1242, 1492</td><td>10</td><td>L3</td><td>CO2</td></tr>
<tr><td>5</td><td>Write a Note on Following
i. Population and Sample
ii. Parameter and Statistics
iii. Descriptive and Inferential Statistics
iv. Probability and Probability distribution</td><td>10</td><td>L2</td><td>CO2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	20	--	--	--	10	40	--	--	--

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Academic year 2023-2024 (EVEN Sem)

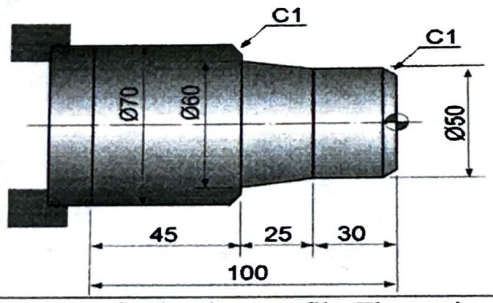
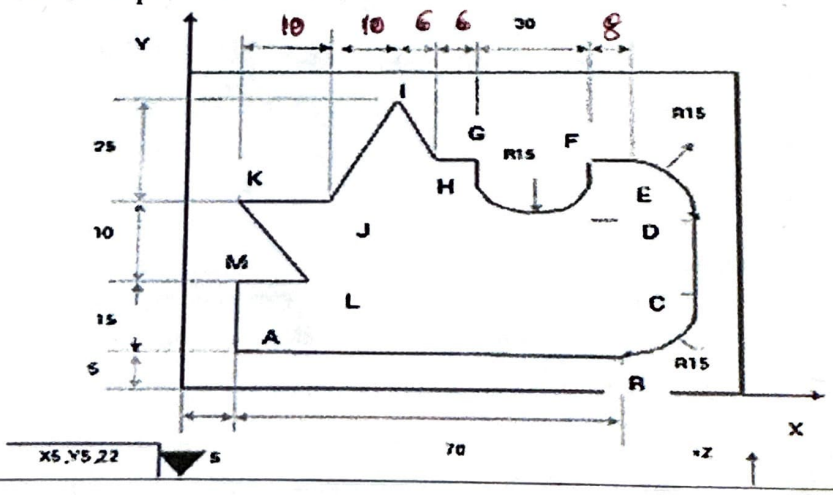
DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18th June 2024**	**Maximum Marks**	**50**
Course Code	**IM343AI**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	
CAD/CAM & ROBOTICS			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	L1	CO1
2	State are the differences between the following? a) absolute and incremental coordinate system b) machine zero and program zero	10	L2	CO1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. C1=3X45^0 	15	L3	CO1
4	Write a CNC Milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5mm. 	15	L3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	50	--	--	--	10	10	30	--	--	--

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th June 2024**	**Maximum Marks**	**50**
Course Code	**IM244AI**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	
OPERATIONS RESEARCH			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	In relation to the LP problem, explain the implications of the following assumptions of the model: i. Linearity of the objective function and constraints, ii. Continuous variables, iii. Certainty.	10	1	1
2	The company manufactures the models A, B and C whose profit contributions per unit are Rs 20, Rs 50 and Rs 70 respectively. The minimum weekly production requirement is 30 units for Model A, 150 units for Model B and 60 units for Model C. Each type of tape recorder requires a certain amount of time to manufacture the component parts for assembly and packaging. Specifically, dozens of Model A units require 6 hours to manufacture, 5 hours to assemble, and 2 hour to package. The numbers corresponding to a dozen units of Model B are 5, 6 and 4 and for dozens of Model C units are 7, 8 and 5. Over the next week, the company provides 150 hours of manufacturing, 140 hours of assembly and 60 hours of packing time And packaging. Formulate this problem as a LP model for maximizing the company's gross profit.	10	2	2
3	Solve the following Linear Programming Problem by Graphical Method: Maximize $Z = 300x_1 + 400x_2$ Subject to the constraints $5x_1 + 4x_2 \leq 200$ $3x_1 + 5x_2 \leq 150$ $5x_1 + 4x_2 \geq 100$ $8x_1 + 4x_2 \geq 80$ where $x_1, x_2 \geq 0$	10	3	3
4	There are four machines and four operators with their respective payoffs (in 100 rupees) as shown below. How the machines should be allocated to each operator so as to minimize the total pay – off. Operators Machines O_1 O_2 O_3 O_4 M_1 5 4 2 7 M_2 7 8 3 5 M_3 5 3 4 6 M_4 4 6 6 5	10	2	2

5	Five salesmen are to be assigned to five districts. Estimates of sales revenue (in thousands) for each salesman are given as follows: Find the assignment pattern that maximizes the sales revenue.	10	3	3

	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**1.07.2024**	**Maximum Marks**	**50**
Course Code	**IM345AT**	**Duration**	**90 Min**
Sem	**IV**	**CIE – I**	

MARKETING MANAGEMENT

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	Describe the key principles of digital marketing and explain how they differ from traditional marketing approaches. Provide examples to illustrate your points.	10	L2	CO1
2	Your company has a limited marketing budget and needs to choose between Search Engine Optimization, Pay Per Click, and Social Media Marketing to promote a new product. Based on the product being a high-tech gadget aimed at young professionals, which channel(s) would you prioritize and why? Justify your decision with specific examples.	10	L3	CO1
3	Identify and explain at least five key metrics used to measure the impact and performance of content marketing efforts. How can these metrics influence future content strategies?	10	L2	CO1
4	Imagine you are a digital marketing strategist at GreenTech Innovations, a company launching a new line of eco-friendly kitchen appliances called "EcoChef." Design a comprehensive content marketing strategy to promote these products. Include specific goals, types of content, distribution channels, and methods of engagement with your target audience.	10	L3	CO1
5	Explain the importance of data-driven insights in social media marketing. Describe the process of developing audience insights using social media analytics tools. How can these insights inform the development and optimization of social media campaigns?	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	**Test**	**Max Marks**	40	10	--	--	--	30	20	--	--	--

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DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	22 June 2024	Maximum Marks	25
Course Code	HS248AT	Duration	60 Mins
Sem	IV	CIE – I	

Universal Human Values

QNo.	Questions	M	BT	CO
1.	Define self-exploration. What is the content of self – exploration?	5M	L3	CO1
2.	"Right understanding + Relationship = Mutual fulfilment; Right understanding + Physical facilities = Mutual prosperity." Illustrate the above with two examples for each.	8M	L3	CO3
3.	How can we ensure harmony in self ('I')?	4M	L1	CO4
4.	How do prosperity and wealth differ? What is more acceptable to us and why?	8M	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	5	8	8	4	4	8	13	-	-	-

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	5th September 2023	Maximum Marks	50
Course Code	21IM41	Duration	90 Min
Sem	IV	Improvement Test	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.
2. Use of statistical tables permitted.

Q. No.	Questions	M	BT	CO
1.	An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approximately normally distributed. A random sample of 12 specimens has a mean tensile strength of 3250 psi. i. Test the hypothesis that mean strength is 3500 psi. Use 1 % significance level. Standard deviation is known to be 60 psi ii. What is the smallest level of significance at which you would be willing to reject the null hypothesis in part (i)? iii. Explain how you could answer the question in part (i) with a two-sided confidence interval on mean tensile strength. iv. How will the test be performed if the standard deviation is not known?	(10)	L3	CO3
2.	Inter arrival times in minutes between successive customers in a hotel is exponentially distributed with a mean of 2 minutes. i. Compute the probability that inter arrival time between two customers is at most 2 minutes. ii. What is the probability that at least 2 customers arrive in one minute? iii. What is the mean-time till every 3rd customer arrives? iv. What is the probability of no customers arriving in a span of 3 minutes?	(10)	L3	CO3
3.	i. What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region for this test? ii. State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45 iii. What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood?	(10)	L2	CO4
4.	For the sample (5,2,3,1,0,0,1,3,3) from a population, i. Construct a 90 % confidence interval on population mean if the population standard deviation is known to be 2. ii. What should the sample size be in order to keep the width of the confidence interval to less than 2? iii. Construct a 95 % confidence interval on population mean if the population variance is unknown.	(10)	L3	CO4
5.	An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find i. the probability that a bulb burns between 778 and 834 hours. ii. Probability that a bulb has a life of at least 1000 hours iii. What percent of bulbs have their life within ± 2σ from their own mean life, where σ is the standard deviation?	(10)	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Test Max Marks	--	--	20	30	--	10	40	--	--	--

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CIE - I - 2023

Academic year 2022-2023 (Even Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**5th July 2023**	**Maximum Marks**	**50**
Course Code	**21IM41**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE-I**	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.

Q. No.	Questions	M	BT	CO
1.	20 weight measurements in a process were recorded as: 61 24 23 27 29 59 47 38 49 12 36 75 64 83 92 18 29 18 20 19 i. Draw a box plot for the data. ii. What are the range and the Inter Quartile Range for the data? iii. Are there any outliers? If yes, what are they? iv. Draw the stem and leaf diagram and comment on the distribution of data	(10)	L3	CO1
2.	Three machines A, B and C produce respectively 50%, 30% and 20% of total number of items in a factory. The proportion of defective output of these machines are 0.03, 0.04 and 0.05 respectively. If an item is selected at random find (i) The probability that the item selected is defective. (ii) If through an action of improvement, the percentage of defective output is reduced to 2%, 2% and 3% respectively. Revise the probability estimate of (i) above.	(10)	L3	CO1
3.	A multiple-choice test contains 24 questions, each with four answers. Assume a student just guesses on each question. (a) What is the probability that the student answers at least 21 questions correctly? (b) What is the probability the student answers at most 10 questions correctly?	(05) (05)	L3	CO2
4.	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) a) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G)	(10)	L3	CO2
5.	A random variables X is distributed as $f(x) = x/k$, $1 \le x \le 4$ a) What is the probability that X will take on a value of at most 2? b) Plot the probability density function of X. c) Derive the cumulative distribution of X and plot the same. d) What is the median of X? Is it the same as mean of X?	(03) (02) (03) (02)	L3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	30	--	--	--	--	50	--	--	--

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Academic year 2022-2023 (Even Sem)

DEPARTMENT OF

BIO TECHNOLOGY

Date	05th July 2023	Maximum Marks	25
Course Code	21BT42A	Duration	60 mins
Sem	IV	CIE 1	
ENVIRONMENTAL TECHNOLOGY			

Instructions to Candidates:

- Answer all the questions, Write the answers in equivalence to mark assigned

Q. No	Question	Marks	CO	BTL
1	State the SDG's promulgated by United Nations.	5	1	2
2	Discuss in brief the different components which constitute the environment.	5	1	3
3	State the significance of environmental education and NGO's in conserving the environment.	5	1	2
4	Explain the step by step process of Environmental Impact Assessment with flow chart.	5	1	2
5	Discuss the objectives and applications of ISO14000 series in Environmental Management System (EMS)	5	1	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	-	-	-	-	20	05	-	-	-

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Academic year 2022-2023 (Even Sem)

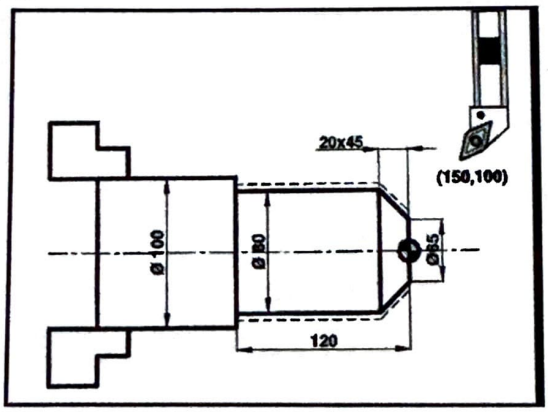
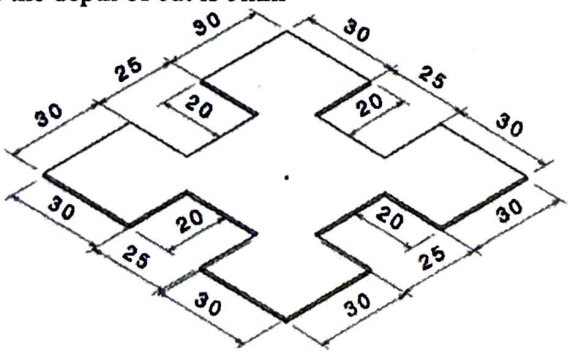
DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**6th July 2023**	**Maximum Marks**	**50**
Course Code	**21IM43**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE - I**	
CAD/CAM & ROBOTICS			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	State the differences between Lathe, Numeric control machine and Computer numeric control with respect to quality, time, cost and automation?	10	L1	CO1
2.	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	L2	CO1
3.	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. 	15	L4	CO2
4.	Write a CNC milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5mm 	15	L4	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	30	--	--	10	10	--	30	--	--

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Academic year 2022-2023 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**6th July 2023**	**Maximum Marks**	**50**
Course Code	**21IM44**	**Duration**	**90 Min**
Sem	**IV**	**CIE – I**	
MARKETING MANAGEMENT			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Explain the concept of market segmentation. Describe the bases for market segmentation.	10	2	1
2.	How digital marketing differs from traditional way of marketing. Highlight five key differences.	10	2	2
3.	How would you measure the performance of content marketing? Can you describe one example of good content you have experienced?	10	3	2
4.	Create a good content for any product of your choice. Take the help of infographics, information and quotes. Content should not exceed one page of your answer booklet.	10	3	3
5.	Analyse the various important elements of "marketing mix" to deliver value to the customer.	10	2	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	20	10	--	--	30	20	--	--	--

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Academic year 2022-2023 (Even Semester)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	07th July 2023	Maximum Marks	25
Course Code	21HSU48	Duration	60 Mins
Sem	IV	CIE – I	

UNIVERSAL HUMAN VALUES

QNo.	Questions	M	BT	CO
1.	Distinguish between happiness and unhappiness?	5	4	3
2.	How to achieve harmony with the body?	5	1	1
3.	What is prosperity? Is it different from happiness?	5	2	1
4.	What is value education? Why there is a need of value education? How does value education helps in fulfilling one's aspirations?	5	1	2
5.	What is the difference between prosperity and wealth? What is more acceptable to us and why?	5	3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	5	5	5	10	5	5	5	--	--